UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Artificial Intelligence

POST-MODULE ASSESSMENT
SEMESTER 1, SESSION 2025/2026

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COURSE CODE	:	**SAIA 2143**
COURSE NAME	:	**DATA WRANGLING AND VISUALIZATION**
PROGRAMME	:	**BACHELOR OF ARTIFICIAL INTELLIGENCE**
DURATION	:	**3 WEEK**
DATE	:	**JANUARY 2026**

INSTRUCTIONS TO CANDIDATES:

1. Answer ALL questions in **Microsoft Word and** upload in e-learning **before 26 January 2026 (11.59 pm).**
2. Save the file in the format:
   StudentName_MatricNumber_CourseCode(section)_LecturerName_Sem1_2526
   (***Example****: Hani_MAN191010_SAIA2143(01)DrNorzihaMegat_Sem1_2526*)
3. Save the answer script file in **Microsoft Word** and **pdf format**.
4. Written PMA should be formatted – **Font: Arial, Size:12, Line spacing: 1.5, Page number: Bottom centre, Max Page: 15-20 page.**
5. Upload files **(Microsoft Word and pdf)** in e-learning and email to your lecturer:
   Section 1 - **norziha.kl@utm.my**
   Section 2 - **septavera@utm.my**
   Section 3 - **doriswong@utm.my**

WARNING!!

Students caught copying/cheating during the examination will be liable for disciplinary actions and the Faculty may recommend the student to be expelled from the study

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This PMA question consists of (10) printed pages only including this page

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Plagiarism

1.0 PLAGIARISM

(Excerpt from Item 27, UTM Graduate Studies Rules and Regulations)

1.1 Plagiarism in academics includes producing thesis, dissertation, project report, article, coursework and research findings without acknowledging or referring to the original sources and claiming as one's own work. It involves the submission of the whole or part of the work towards a degree.

1.2 A Student who committed plagiarism will be penalized based upon decision made by the Faculty Academic Committee.

1.3 Details of prohibition against plagiarism can be referred to UTM Student Regulations Book.

2.0 EXAMINATION MISCONDUCT

2.1 Students are **NOT** allowed to do the following during the examination:

a) Give/receive/own any electronic, printed or other forms of information that is not now allowed during the examinations whether it be inside or outside the examination hall unless with the permission of the chief invigilator

b) Make use of the above information to answer the examination questions.

c) Cheating, attempt to cheat or act in manner that is construed as cheating during an examination.

d) Other misconduct as stated by the university.

3.0 PUNISHMENT

3.1 If a student has acted against the rules and regulations of the examination and is found guilty by the Faculty Examination Committee, the Senate can take any of the following actions:

a) Give a mark of ZERO (0) for the course (including coursework); or

b) A mark of ZERO (0) for all the courses registered in that semester.

Students who have found to act against the rules and regulations of the examinations for the second time will be dealt according to the Universities and Colleges Act, 1971, Procedures according to the University Technology Malaysia (Students Code of Conduct), 1999.

Quote from:

"ACADEMIC GUEDELINE BACHELOR'S DEGREE PROGRAMS FULL-TIME STUDY "

Section A- Answer all questions [100 Marks].

Course Learning Outcome 2: Construct data transformation using data wrangling and visualization techniques to clean and preprocess raw datasets, addressing missing values, outliers, and inconsistencies for analytics.
Taxonomies Graduate Attribute: PLO3, Psychomotor Level 4, IT

Case Study A: Data Cleaning with Excel

[20 Marks]

Title: Improving Data Quality for a Customer Database

Dataset: **CaseA_CustomerData.xlsx**

Scenario:
You are given a messy customer dataset in date formats, missing emails, and duplicated records. Clean this data using Excel tools.

Tasks:

1. Construct a "Conditional Formatting" rule or use the "Go To Special" command to physically highlight all blank cells in the dataset with a yellow fill color.
[4 marks]

2. Display the "Remove Duplicates" operation by selecting the dataset range and configuring the tool to check only the "Name" and "Email" columns before executing the deletion.
[4 marks]

3. Manipulate the date column settings by selecting the cells and applying a custom number format code (dd/mm/yyyy) to unify the visual presentation of all dates.
[4 marks]

4. Construct a Conditional Formatting rule using the "Text that Contains" option to automatically color cells green where the Region column equals "Central".
[4 marks]

5. Assemble a Pivot Table by dragging and dropping the "Region" field into the Rows area and the "Customer Name" field into the Values area to generate a summary count.
[4 marks]

Case Study B: Data Wrangling & Visualization in Python

[40 Marks]

Title: Study on Employee Productivity Levels

Dataset Name: **employees_productivity_dataset.xlsx**

Scenario:
You are a data analyst in the HR Analytics department. The management wants insights into how working hours, remote-work preference, digital tool usage, job satisfaction, and experience level influence employee productivity scores. The dataset contains employee records with variables such as:

- Weekly working hours
- Overtime frequency
- Remote vs Office work preference
- Tool usage proficiency
- Job satisfaction score
- Years of experience
- Productivity score

Answer/complete the following tasks, using Python. Please include screenshots of the Collab code and results, as well as your answer/ analysis/ insight:

1. Using appropriate Phyton command**,** display the dataset information including data types and non-null counts.

[2 marks]

2. Using Python grouping operations, construct a grouped summary table by *Remote Work Preference* that displays:
   a. Count of employees in each group
   b. Mean productivity score
   c. Average weekly working hours

   Then produce a short written interpretation based on this, for example, which group has the highest productivity? Does longer hours always lead to higher productivity?

[4 marks]

3. Using Seaborn, create a box plot with the following parameters, and the observe and state one operational insight you can obtain from it:
   - X-axis: Remote Work Preference
   - Y-axis: Productivity Score
   - Hue: Job Satisfaction Level (e.g., Low/Medium/High)
   - Style: Apply seaborn's "whitegrid" theme

[10 marks]

4. By visually inspecting the generated box plot in the previous answer, identify which work preference group demonstrates the highest median productivity score.

[2 marks]

5. Using bar charts, construct and compare visualisations that show how (a) job satisfaction and (b) tool proficiency may affect the productivity of employers.

[8 marks]

6. Using Python, construct a stacked bar chart to compare how average job satisfaction differs across departments.

[4 marks]

7. Using Plotly, build an interactive scatter plot to visualize the relationship between:
   - X-axis: Years of Experience
   - Y-axis: Productivity Score

   a. Color: Remote Work Preference

   [6 marks]

   b. Enable at least ONE interactive feature (hover info, slider, or dropdown).

   [2 marks]

   c. State TWO interpretation or insights obtained from the chart .

   [2 marks]

Case Study C: Dashboard with Google Looker Studio

[30 Marks]

Title: Global Climate Monitor: Weather Extremes & Air Quality Dashboard

Submission Format: PDF Document containing **Screenshots (Print Screens)** of your Google Looker Studio Dashboard.

Scenario: You have been appointed as a Climate Data Analyst for the *Global Environmental Watch*. With the upcoming International Climate Summit approaching, the committee has provided you with a dataset (`GlobalWeatherRepository.csv`) containing weather data from 211 countries.

Your objective is to build a real-time monitoring dashboard to track **Temperature Anomalies** and **Air Pollution Levels**. You must capture screenshots of your findings to present in the summit's opening report.

Question 1: Temperature & Weather Patterns

The committee needs to identify the hottest regions and understand how "feels like" temperature differs from reality.

a) Construct a new Calculated Field by entering the formula editor and typing the syntax to determine the `Heat Variance`.
Formula Hint: Calculate the difference between what it feels like and the actual temperature: `feels_like_celsius - temperature_celsius`.

Submission: Print Screen of your formula editor window showing the formula.
[4 Marks]

b) Assemble the Geo Chart (Map) using `country` as the dimension and Max `temperature_celsius` as the metric.

Submission: Print Screen of the map. Which specific country appears as the hottest (darkest color) on the map?
[4 Marks]

c) Manipulate the chart settings of a Bar Chart to display `condition_text` vs `temperature_celsius`. Calibrate the sorting order to "Descending" based on the metric to physically rearrange the bars from highest to lowest.

Submission: Print Screen of the sorted bar chart. Which specific weather condition ("Sunny", "Mist", etc.) has the highest average temperature?
[4 Marks]

d) Insert a Scorecard component onto the canvas and set the metric to `wind_kph`. Adjust the aggregation setting to "Average".

Submission: Print Screen of the scorecard showing the value.

[3 Marks]

Question 2: Air Quality & Health Safety

Health officials are concerned about air pollution (PM2.5) and UV radiation exposure.

a) Construct a data in a Table by adding `country` as the Dimension and `air_quality_PM2.5` as the Metric. Arrange the rows by clicking the header or setting the sort property to display the highest pollution values at the top.

Submission: Print Screen of the table. Which country has the highest average PM2.5 level?

[4 Marks]

b) Plot a Scatter Chart by assigning `temperature_celsius` to the X-Axis and `humidity` to the Y-Axis. Adjust the chart size to ensure the spread of data points is clearly visible.

Submission: Print Screen of the chart. Is the relationship between temperature and humidity **Positive** (both go up) or **Negative** (one goes up, one goes down)?

[4 Marks]

c) Execute a filter setup on a Scorecard. Navigate to the "Setup" tab, click "Add a Filter," and configure the condition to include only records where `uv_index` is greater than (>) 5.

Submission: Print Screen of the scorecard. How many records have a "High" UV index (>5)?

[4 Marks]

d) Display a Pie Chart on the canvas. Select `condition_text` as the dimension and ensure the chart segments are automatically generated based on the Record Count.

Submission: Print Screen of the chart. What is the #1 most common weather condition in this dataset?

[3 Marks]

Description: This dataset contains real-time weather and air quality data from 211 different countries.

1. Location & Time

*These columns tell us **where** and **when** the weather data was recorded.*

- **country**: The name of the country (e.g., "Malaysia", "Japan", "USA").
- **location_name**: The specific city or region within that country (e.g., "Kuala Lumpur", "Tokyo").
- **latitude**: The geographic coordinate specifying North/South location.
- **longitude**: The geographic coordinate specifying East/West location.
- **timezone**: The local time zone of the location (e.g., "Asia/Kuala_Lumpur").
- **last_updated**: The exact date and time the weather was recorded (YYYY-MM-DD HH:MM).
- **last_updated_epoch**: The time recorded in "Unix format" (total seconds since Jan 1, 1970). *Mainly used by computers for sorting.*

2. Temperature & Weather Condition

*These columns describe **how hot/cold** it is and what the sky looks like.*

- **temperature_celsius**: The current air temperature in degrees Celsius (°C).
- **temperature_fahrenheit**: The current air temperature in degrees Fahrenheit (°F).
- **feels_like_celsius**: The "Real Feel" temperature. This accounts for humidity and wind (e.g., it might be 30°C but "feel like" 35°C due to humidity).
- **feels_like_fahrenheit**: The "Real Feel" temperature in Fahrenheit.
- **condition_text**: A text description of the weather (e.g., "Sunny", "Partly Cloudy", "Heavy Rain").
- **uv_index**: A number from 1-11+ measuring the strength of sunburn-causing ultraviolet radiation.
    - *1-2 (Low), 11+ (Extreme).*

3. Wind & Atmosphere

*These columns describe the **movement of air** and **storm conditions**.*

- **wind_kph**: Wind speed in Kilometers per Hour (km/h).
- **wind_mph**: Wind speed in Miles per Hour (mph).
- **wind_degree**: The direction the wind is coming from, measured in degrees (0° = North, 90° = East).
- **wind_direction**: The compass direction of the wind (e.g., "N", "SW", "ENE").
- **gust_kph**: The speed of sudden, brief bursts of wind (gusts) in km/h.
- **gust_mph**: The speed of wind gusts in mph.
- **pressure_mb**: Atmospheric pressure in millibars. Lower pressure often indicates a storm is coming.
- **pressure_in**: Atmospheric pressure in inches of mercury (standard US measurement).

4. Rain, Visibility & Clouds

*These columns describe **moisture** and **clarity**.*

- **precip_mm**: The amount of precipitation (rain/snow) that has fallen recently, in millimeters.
- **precip_in**: The amount of precipitation in inches.
- **humidity**: The percentage (%) of moisture in the air. 100% usually means rain or fog.
- **cloud**: Cloud cover percentage (0% = Clear Sky, 100% = Totally Overcast).
- **visibility_km**: How far you can see clearly, in kilometers. Lower numbers mean fog or haze.
- **visibility_miles**: How far you can see clearly, in miles.

5. Air Quality (Pollution)

*These columns measure **pollutants** in the air. Higher numbers = Worse air quality.*

- **air_quality_PM2.5**: Tiny particles (dust/smoke) small enough to enter the lungs. This is a key measure of pollution.
- **air_quality_PM10**: Larger dust/smoke particles.
- **air_quality_Carbon_Monoxide**: Level of CO gas (often from cars/exhaust).
- **air_quality_Ozone**: Level of Ozone gas (O3).
- **air_quality_Nitrogen_dioxide**: Level of NO2 (often from burning fuel).
- **air_quality_Sulphur_dioxide**: Level of SO2 (often from factories).
- **air_quality_us-epa-index**: A summary score from 1 (Good) to 6 (Hazardous) used by the US EPA.
- **air_quality_gb-defra-index**: A summary score from 1 (Low) to 10 (Very High) used by the UK government.

6. Astronomy (Sun & Moon)

*These columns track the cycle of the **day and night**.*

- **sunrise**: The time the sun rises (e.g., "06:30 AM").
- **sunset**: The time the sun sets (e.g., "07:15 PM").
- **moonrise**: The time the moon rises.
- **moonset**: The time the moon sets.
- **moon_phase**: The shape of the moon (e.g., "Full Moon", "Waxing Crescent").

moon_illumination: The percentage of the moon's face that is lit up (0% = New Moon, 100% = Full Moon).

Case Study D: Ethics in Data Visualization

[10 Marks]

Title: Detecting Bias in Dashboard

Scenario:
You are provided with screenshots of dashboards show in Figure 1. Analyse whether the use of scales, colors, and chart types leads to misinterpretation or bias.

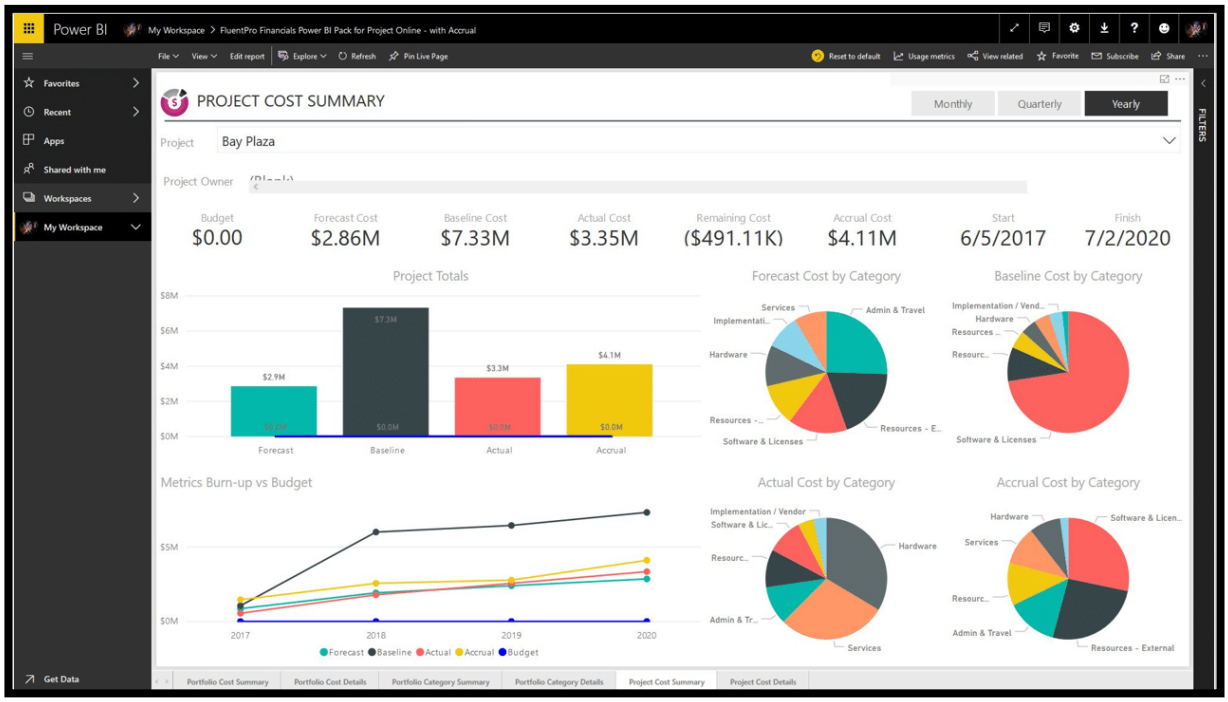


Figure 1: Project Cost Summary Dashboard

Tasks:

1. **Identify** the dashboard and **explains** TWO (2) ethical concerns related to bias, distortion, or misrepresentation.

[4 marks]

2. **Constructs** TWO (2) strategies to enhance ethical clarity and reduce bias in the dashboard.

[4 marks]

3. **Explain** the role of ethical considerations in data communication and support with rationale.

[2 marks]